



UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Hipert/Lab
High Performance Real Time
Lab

1 2 3 4 5 6 7 8 9 10

TAGHeuer

UNIMORE
RACING



About me

Alessandro Toschi

- PhD student in Computer and Data Science @ UNIMORE (started in 2022)
 - My research topic: “*Vehicle motion planning and control in complex environments*”
- UNIMORE Racing team member since 2021

Academic background:

- BS in Automation Engineering @ UNIBO & Tongji University
- MSc in Advanced Automotive Electronic Engineering @ MUNER

The Lab



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High Performance
Real Time **Lab**

- ✓ Complex **workload-intensive** tasks
 - Perception, planning, ML/DNN
- ✓ **Latency-critical** control tasks
 - Cyber-physical interaction
 - Tight actuation loops

<https://hipert.unimore.it/>

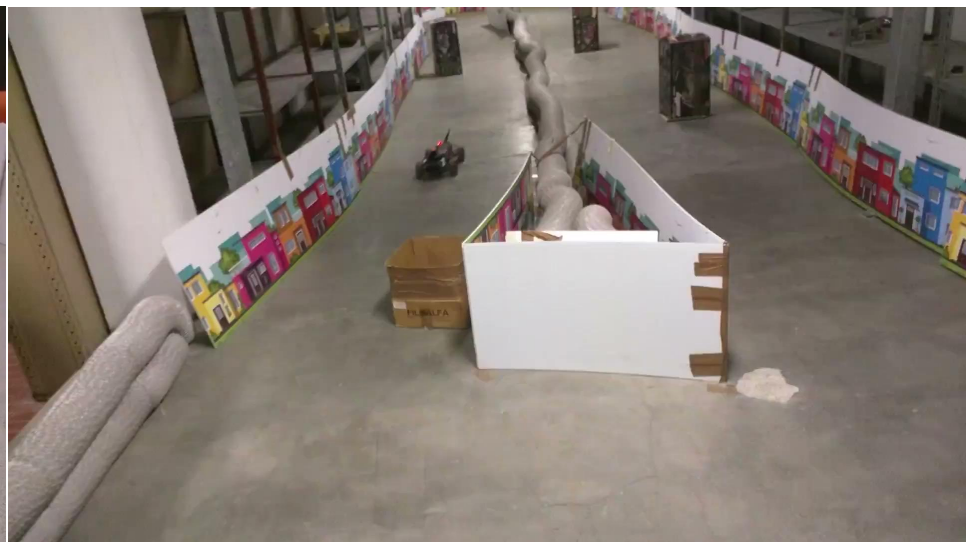


Embedded boards

meets Automotive/Racing



Roboracer (F1tenth)



[Roboracer](#)

Autonomous Racing



Indy Autonomous Challenge Abu Dhabi Autonomous Racing League

by Energy Systems Network (USA) << >> by ASPIRE (UAE)

Since 2021 << >> Since 2024

Indy Light Dallara AV-24 << >> Super Formula Dallara EAV-24

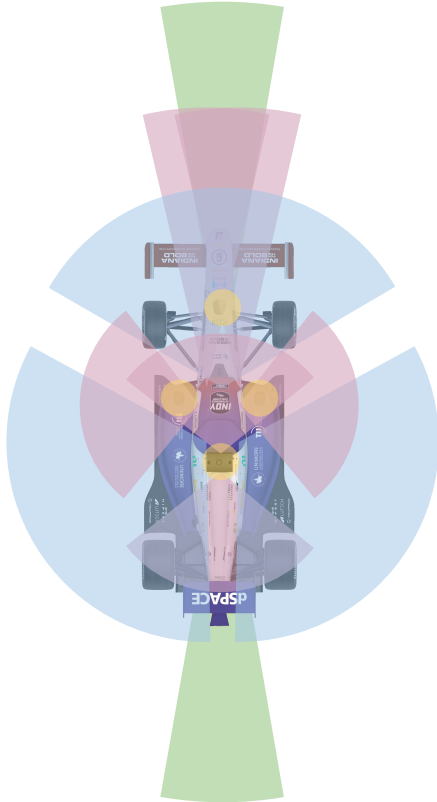
1M\$ first prize << >> 2.25M\$ overall price

1st track: Indianapolis Motor Speedway << >> 1st track: Yas Marina Circuit

Sensors



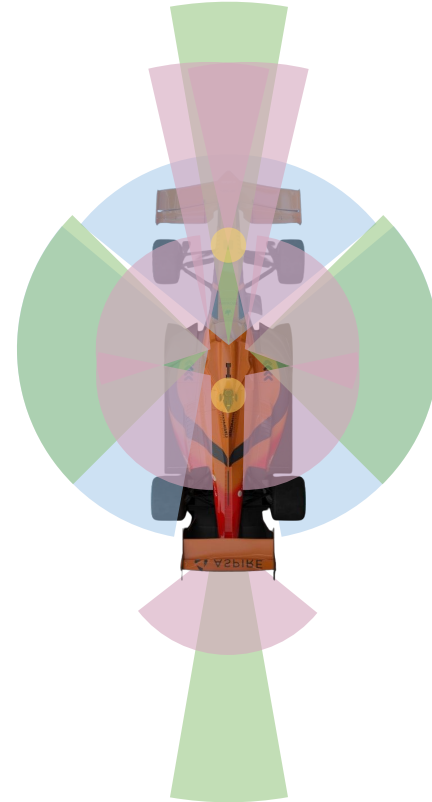
3x 3D Lidars
6x Cameras GS
3x GNSS w RTK
2x 2D radar
phonic wheels



- Lidar
- Camera
- Radar
- GNSS



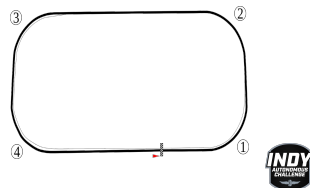
3x 3D Lidars
7x Cameras RS
1x GNSS w RTK
4x 4D radar
1x Optical
sensor
phonic wheels



Racing Events

Indianapolis Motor Speedway (IMS)

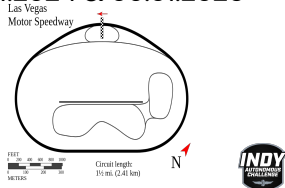
23.10.2021 & 06.09.2024



Fastest lap + 2nd place
2nd place in 2-vehicle race

Las Vegas Motor Speedway (LVMS) - CES

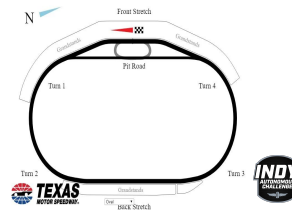
07.01.2022 & 07.01.2023 &
11.01.2024 & 09.01.2025



3rd place
1st place 4-vehicle race
exhibition

Texas Motor Speedway (TMS)

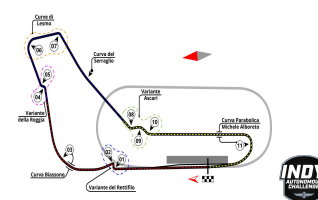
11.11.2022



3rd place

Autodromo di Monza

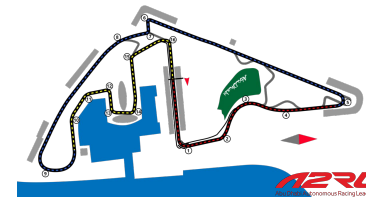
16-18.06.2023



3rd place

YAS marina circuit

25-27.04.2024



2nd place at time trials
3rd place in the race

High Speed - More than 280 km/h



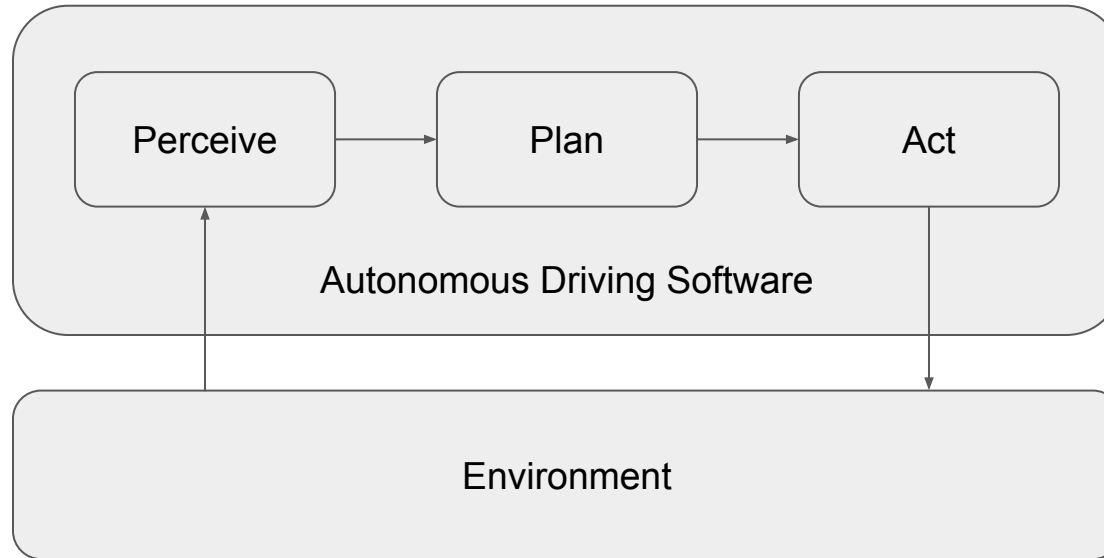
Road Course



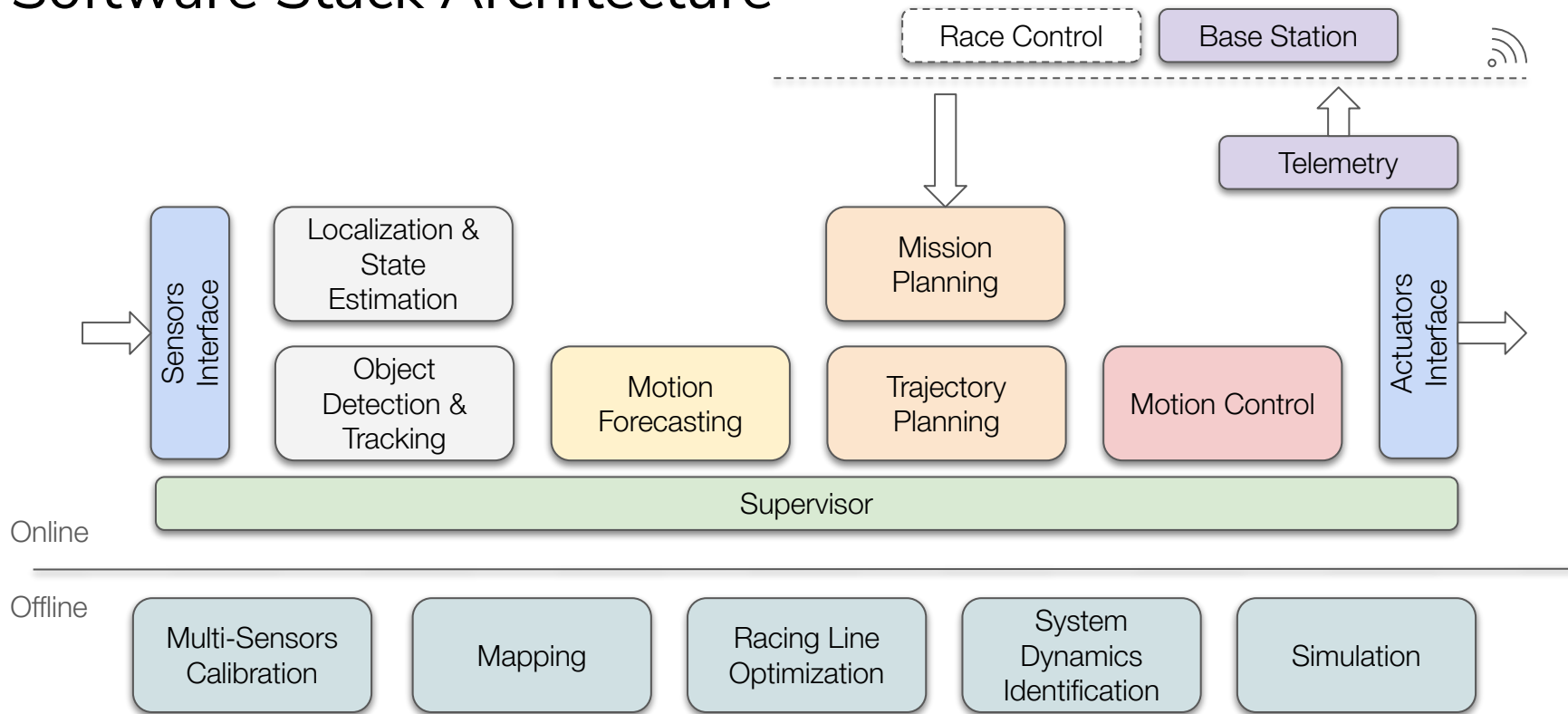
Head to Head Racing



Autonomous Driving Context



Software Stack Architecture



State Estimation

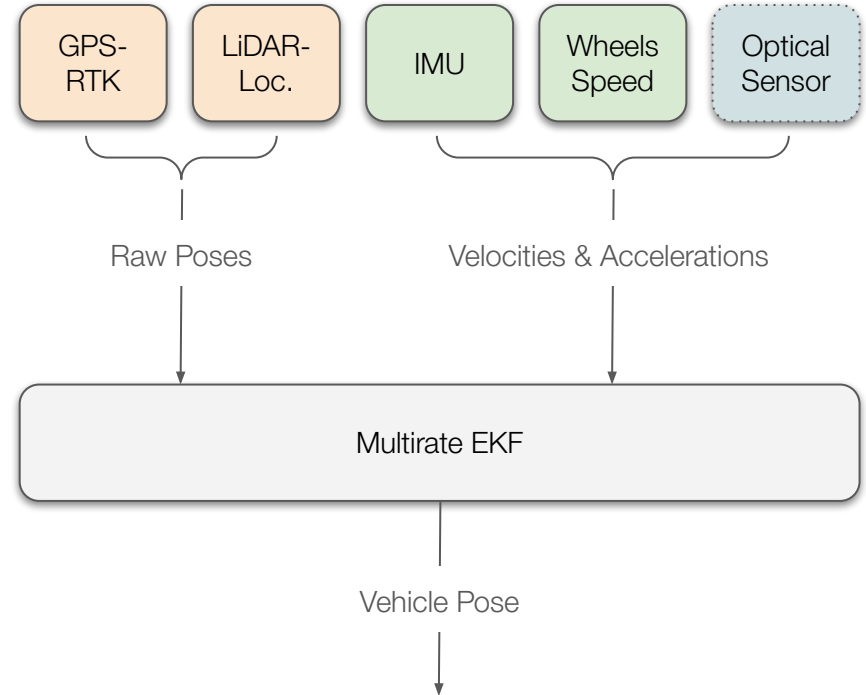
Localization & State Estimation

Centimeter-level accuracy position sources:

- GPS-RTK.
- LiDAR-Localization.

Pose Estimation (EKF):

- Outlier Pre-Filtering.
- Multirate corrections and delay compensation.
- Kineto-Dynamic Bicycle Model.



Side Slip Angle Estimation

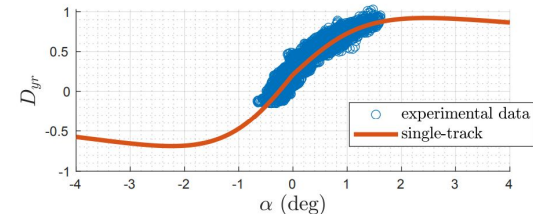
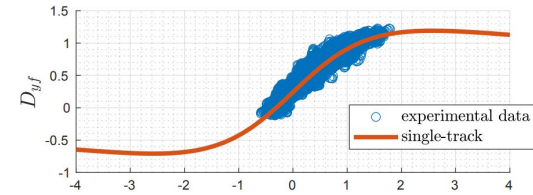
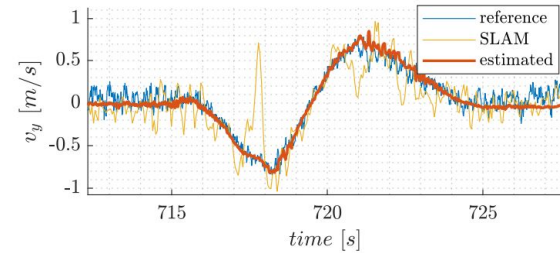
Side Slip Angle Estimation:

- Kalman Filter with Pacejka tire model
- Lateral velocity corrections from LiDAR Odometry.
- WIP: use also Radar odometry

Approaches:

1. Started with Unscented Kalman Filter
2. Now using an Extended Kalman Filter
3. WIP: Moving Horizon Estimation

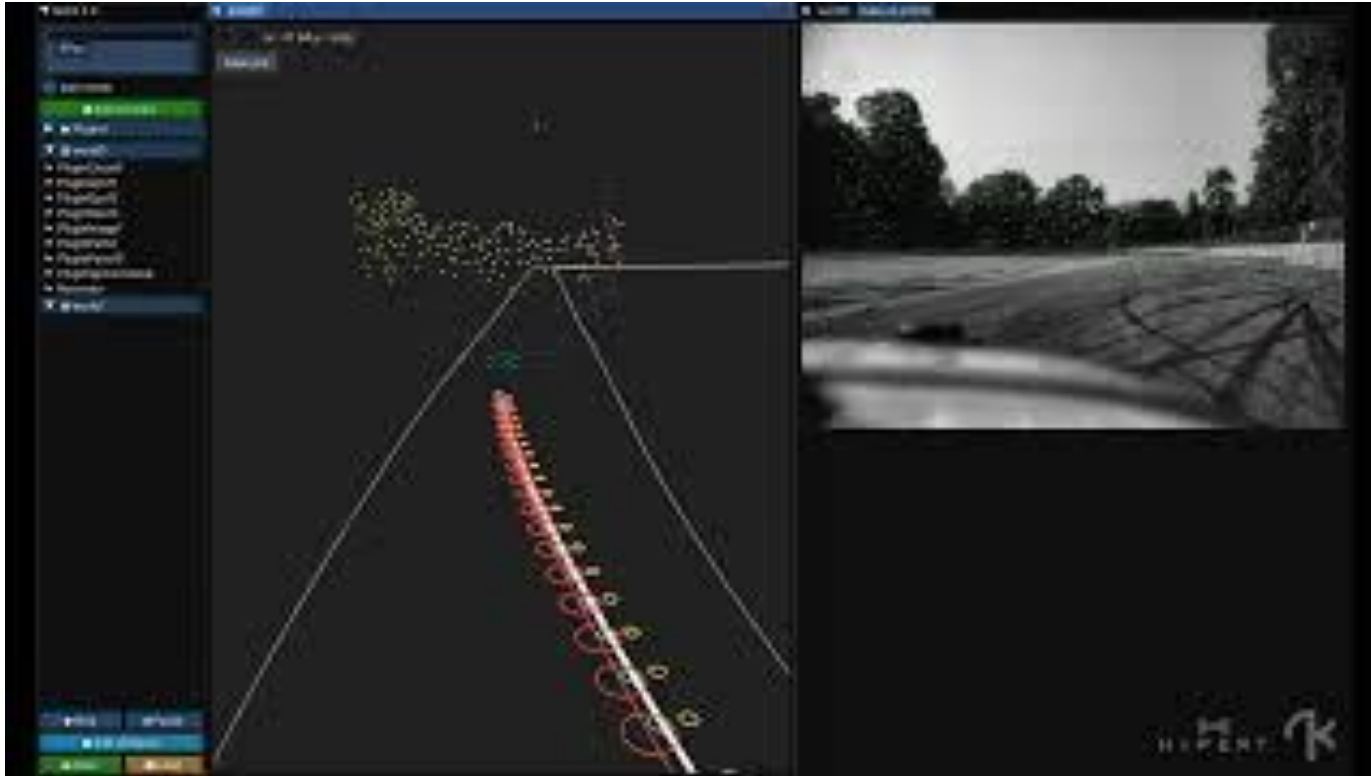
Lateral velocity and tires model parameter estimation.



Lidar Localization



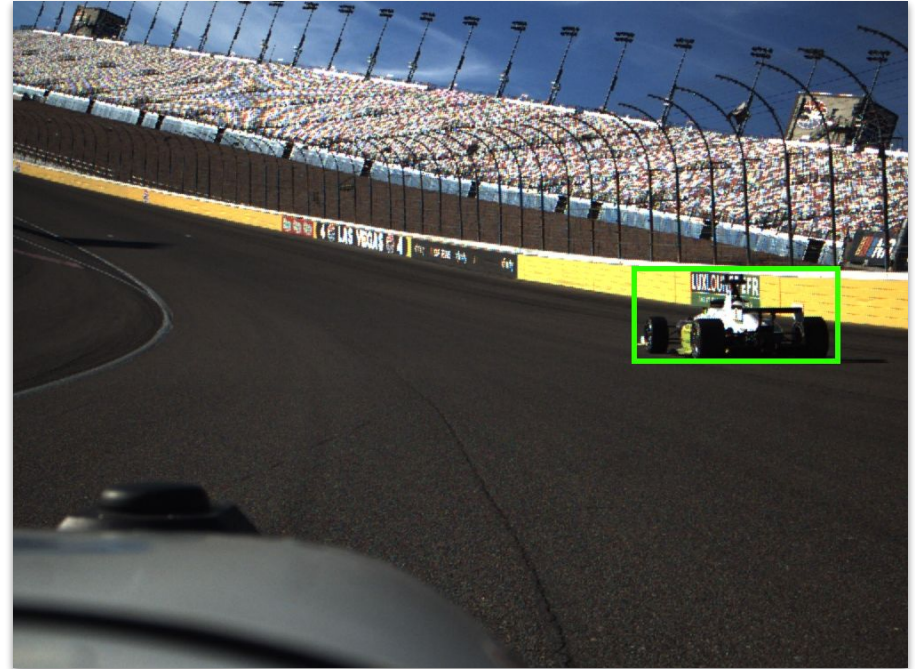
Visual-Inertial Odometry



Perception

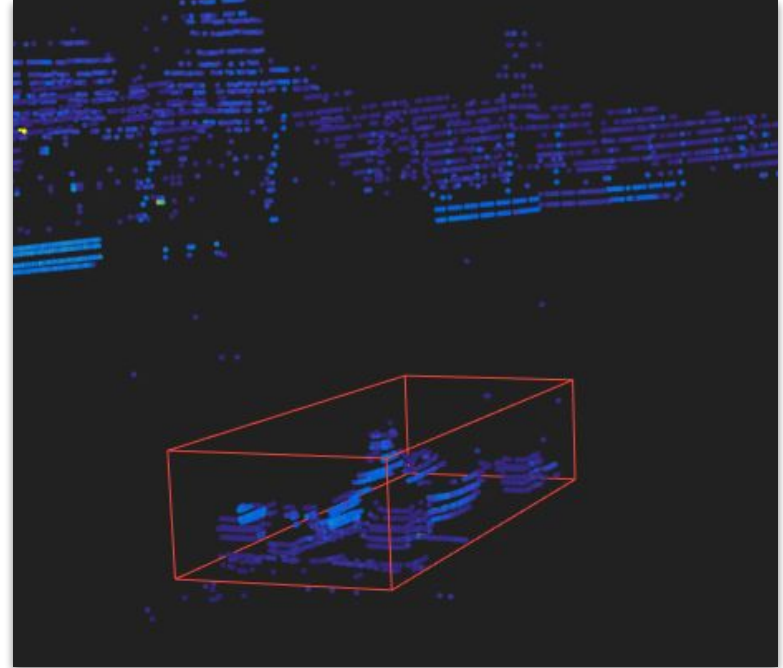
Object Detection - Cameras

- Object detection with YOLOv4
- Optimized inference on tkDNN2
- ~240 FPS on GPU (6 streams)
- Frequency used: 20Hz
- Range: ~ 200 m



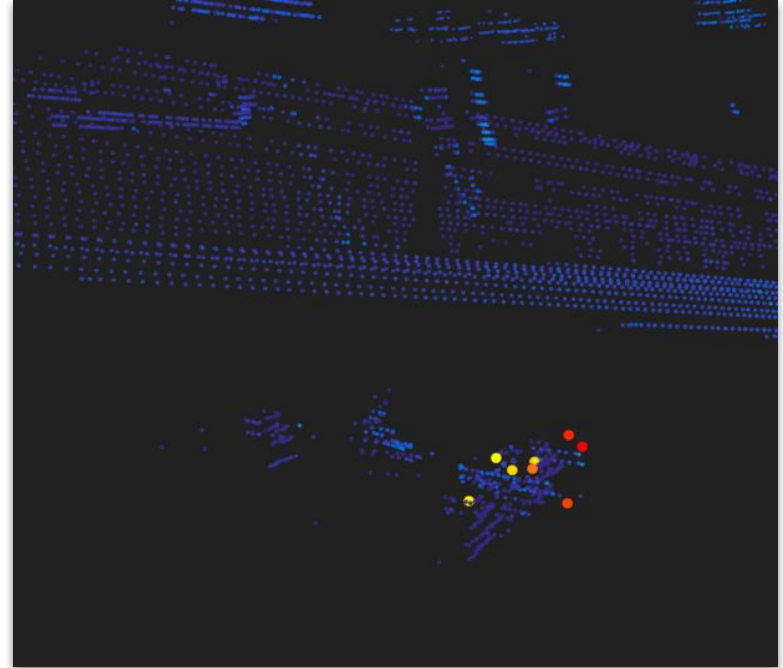
Object Detection - Lidars

- Ground removal
- Object detection with PointPillars
- Optimized inference on tkDNNv2
- ~50 FPS on GPU
- Frequency used: 20 Hz
- Range: ~130 m



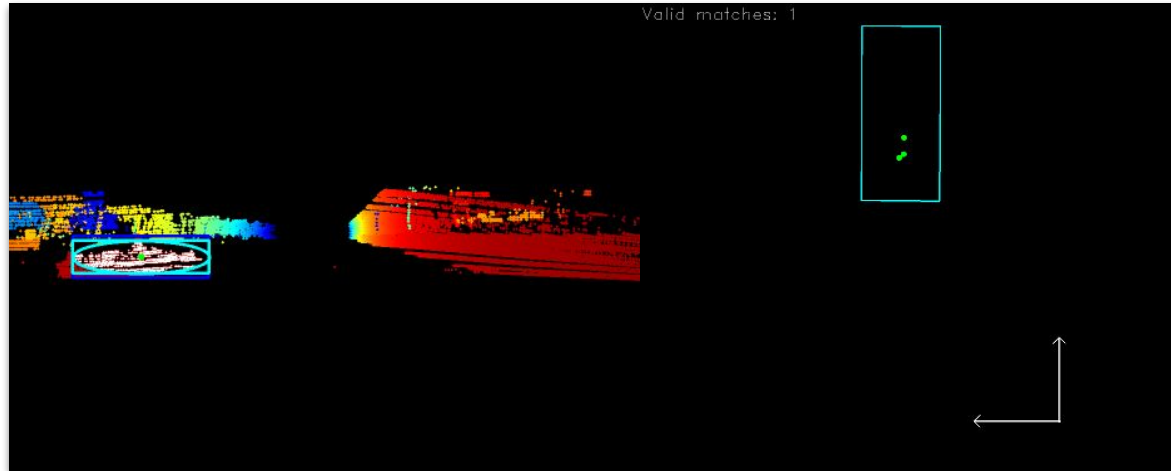
Object Detection - Radars

- Filtering and clustering
- ~20 FPS on CPU
- Frequency used: 20 Hz
- Range: ~150 m



Detections Fusion

- Based on precise multi-sensors calibration
- Uses consensus from multiple sensor to correctly detect an opponent
- Robustness to failures of multiple sensors



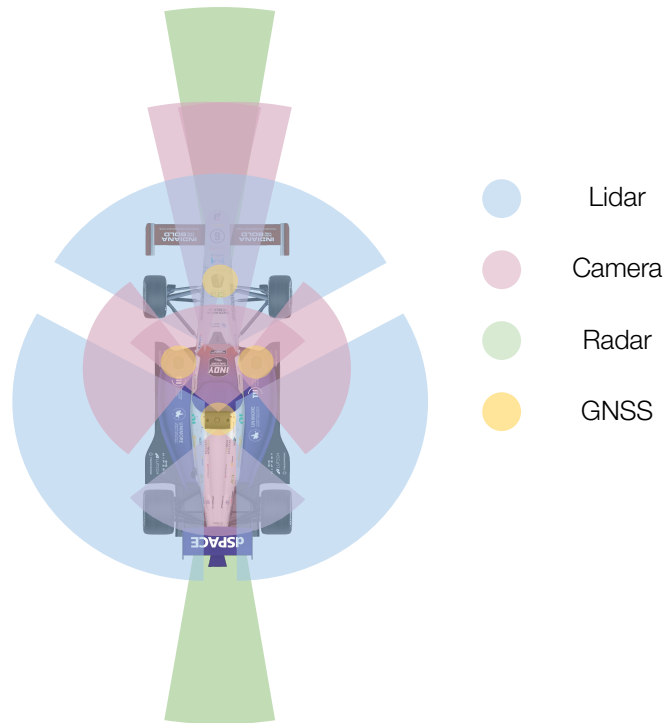
Object Detection & Tracking

Sensor Fusion:

- LiDAR: PointPillars¹.
- Camera: YOLOv4².
- RADAR: Point cloud processing and clustering.

Object Tracking:

- Motion prediction and matching.



[1] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, "PointPillars: Fast Encoders for Object Detection From Point Clouds," in 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, CA, USA: IEEE, Jun. 2019, pp. 12689–12697. doi: 10.1109/CVPR.2019.01298.

[2] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal Speed and Accuracy of Object Detection," Apr. 23, 2020, arXiv: arXiv:2004.10934. doi: 10.48550/arXiv.2004.10934.

Sensors set for the IAC AV-24.

Perception

Sensor Fusion:

- LiDAR: PointPillars.
- Camera: YOLOv4.
- RADAR: Point cloud processing and clustering.

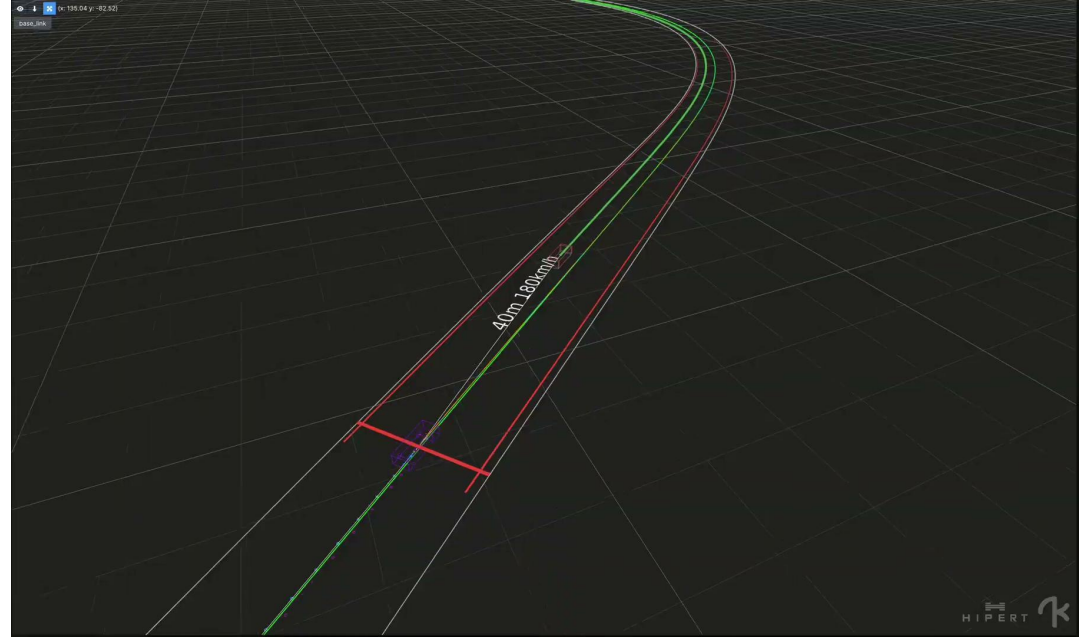
Object Tracking:

- Motion prediction and matching.

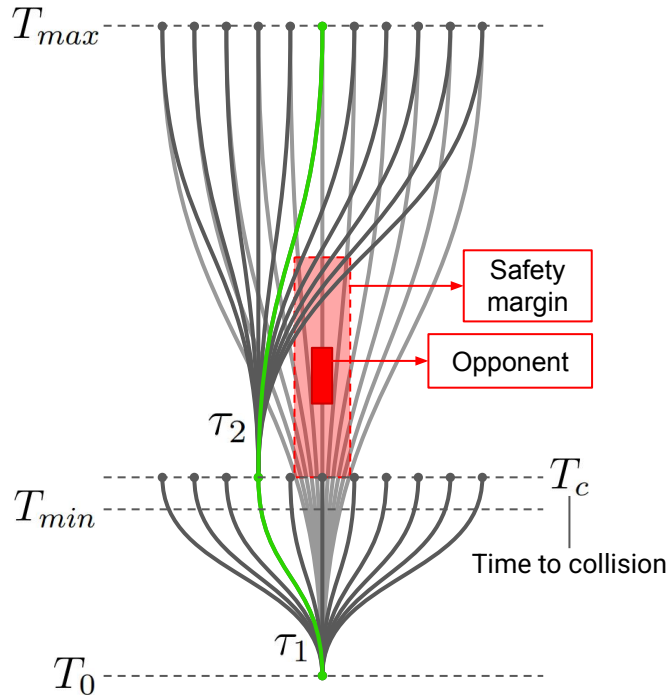


Object Tracking & Forecasting

- Tracker: used to track the opponent position and velocity
- Forecasting: predict the opponent action for the next 6 seconds
- Motion Forecasting methods:
 - Kalman Filter
 - Frenet Planner (currently used)
 - Neural Network



Frenét Graph-Based planner



Generation of a set of trajectories

Cost assignment to each trajectory

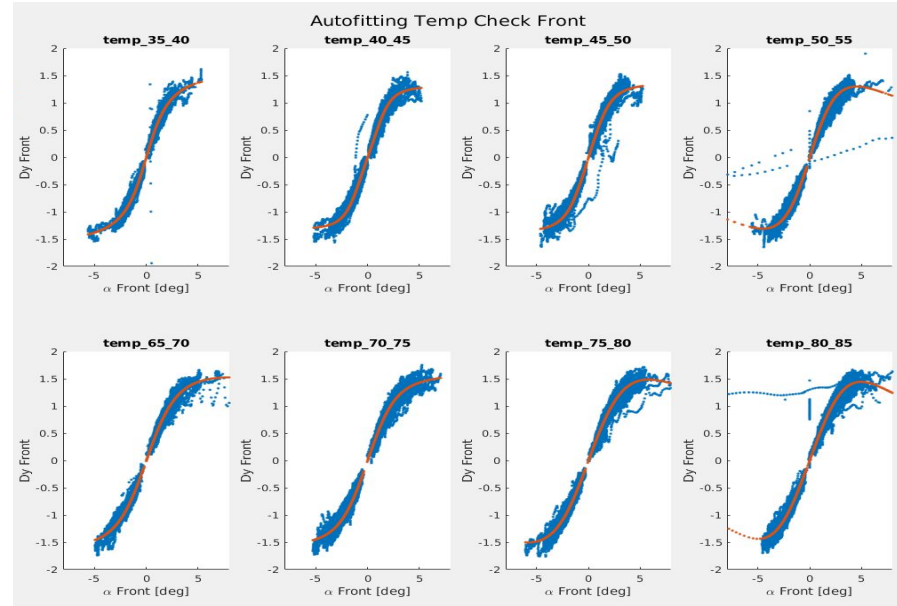
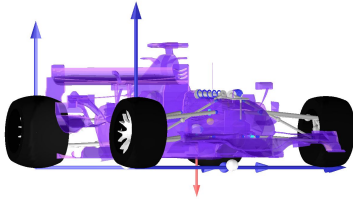
Trajectories collision check

Trajectory selection

Vehicle Dynamics

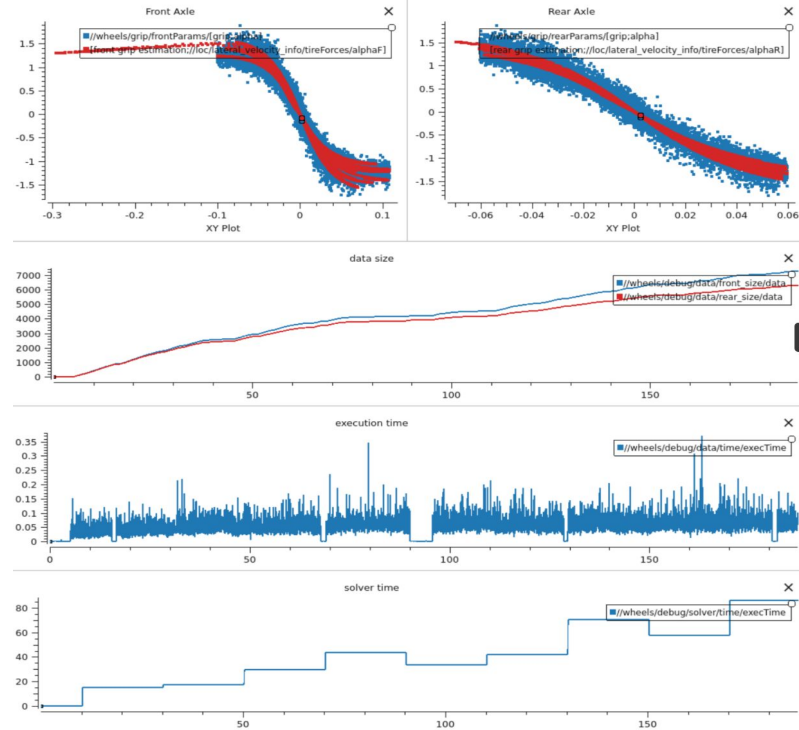
Vehicle Dynamics - Simulations

- High-Fidelity Multibody Modeling and Simulation (Dymola)
- Simulation Environment for Autonomous Software testing (C++, Python, Simulink)
- Automated Batch Simulations with Extensive Boundary and Initial Conditions.



Vehicle Dynamics - Grip Estimation

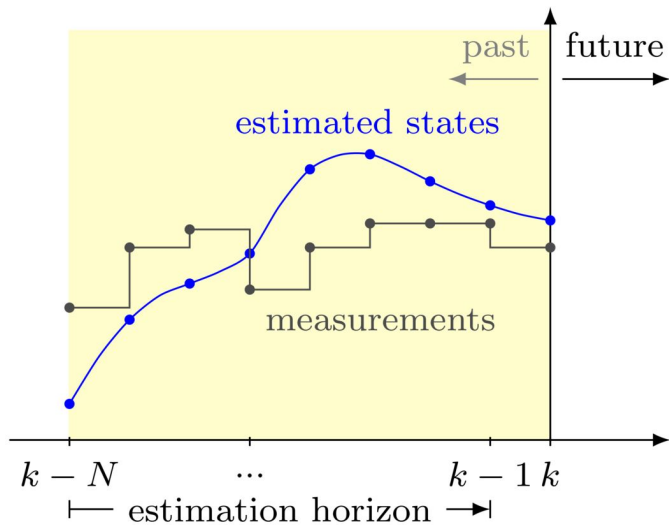
- Real-Time Tire-Road Grip Estimation
- Flexible and Modular Tire Formulations
- Online Parameter Adaptation for Autonomous Controller
- Non-linear regression (with Ceres solver) using estimated vehicle slips



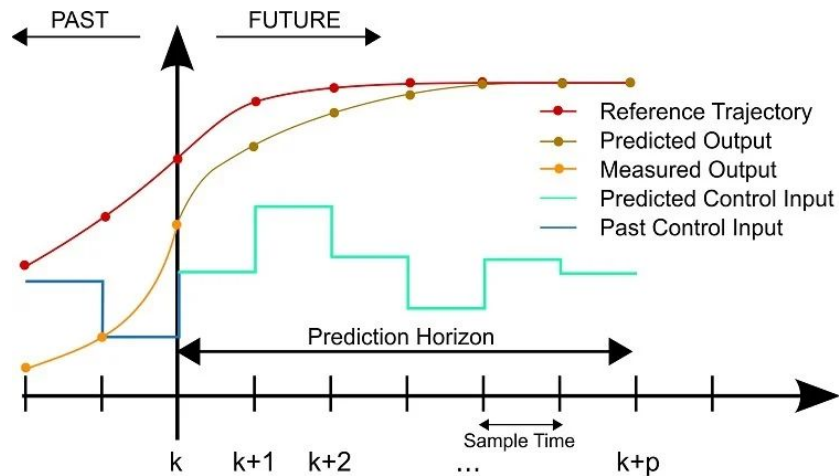
Moving Horizon Optimization

Moving Horizon Optimization

Moving Horizon Estimation



Model Predictive Control



MPC Problem Formulation

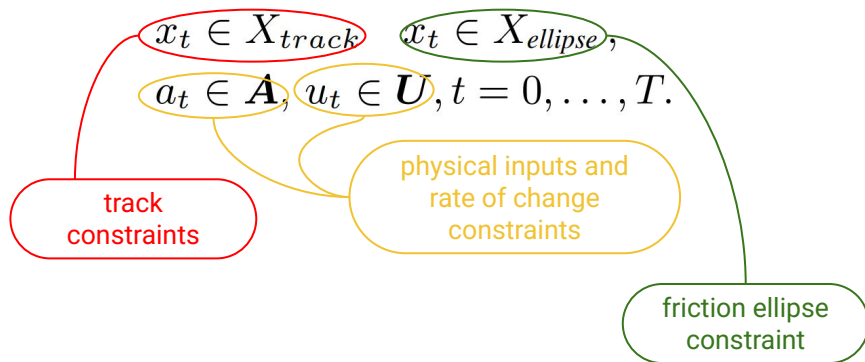
$$\min_{X,U} \sum_{t=0}^T J_{MPC}(x_t, u_t)$$

$$s.t. \quad x_0 = \hat{x},$$

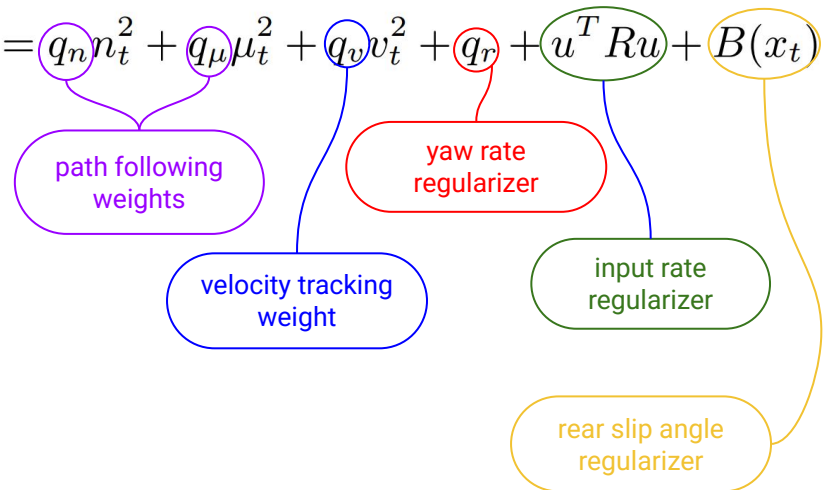
$$x_{t+1} = f_t^d(x_t, u_t),$$

$$x_t \in X_{track} \quad x_t \in X_{ellipse},$$

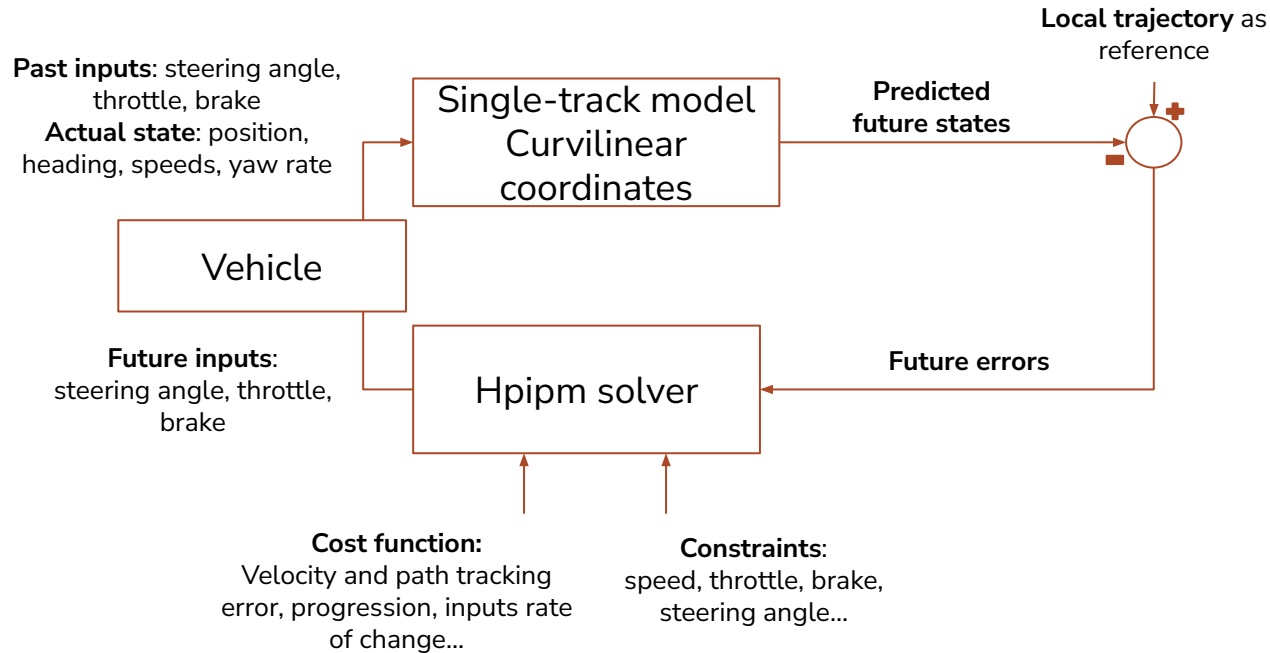
$$a_t \in \mathbf{A}, u_t \in \mathbf{U}, t = 0, \dots, T.$$



$$J_{MPC}(x_t, u_t) = q_n n_t^2 + q_\mu \mu_t^2 + q_v v_t^2 + q_r + u^T R u + B(x_t)$$



Planner and Controller - Model Predictive Control

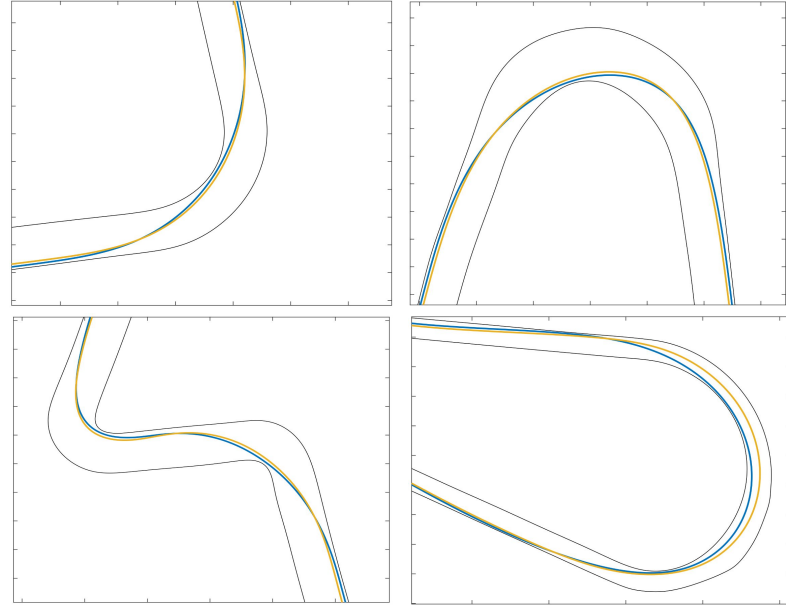


- Dynamics discretized and linearized in time
- **Time horizon of**
 - **2.6 seconds - Controller**
 - **6 seconds - Planner**
- **Discretization and sampling at 40 ms**
- Execution at
 - **100Hz - Controller**
 - **20 Hz - Planner**
- **Quadratic problem** solved using HPIPM (High-Performance Interior Point Method) solver and using CppADCodeGen for automatic differentiation.

Motion Planning and Decision Making

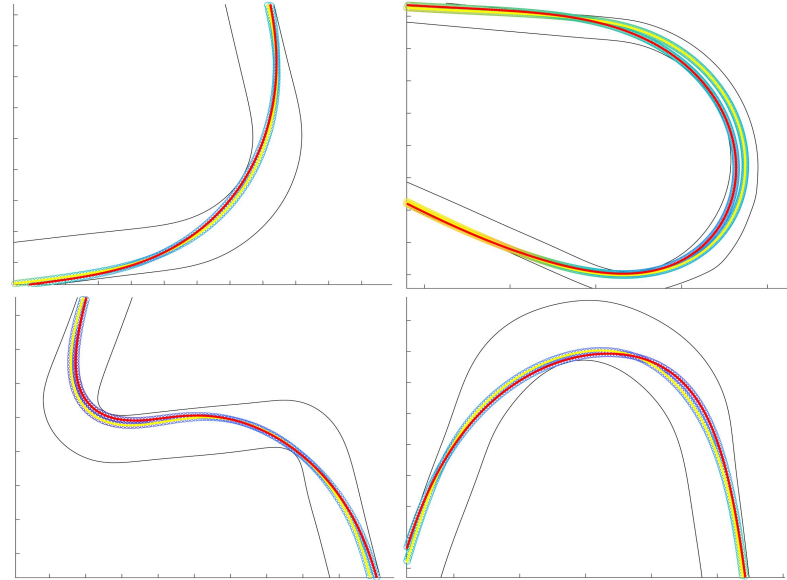
Global Planning - Lap-time Optimization

- Optimal Global Trajectory for Minimum Lap-Time: Optimization in Julia using a single/double track model
- Strategy-Aware Lap-Time Minimization
- Driving Style Comparison
- Lap-Time Sensitivity to Vehicle Parameters



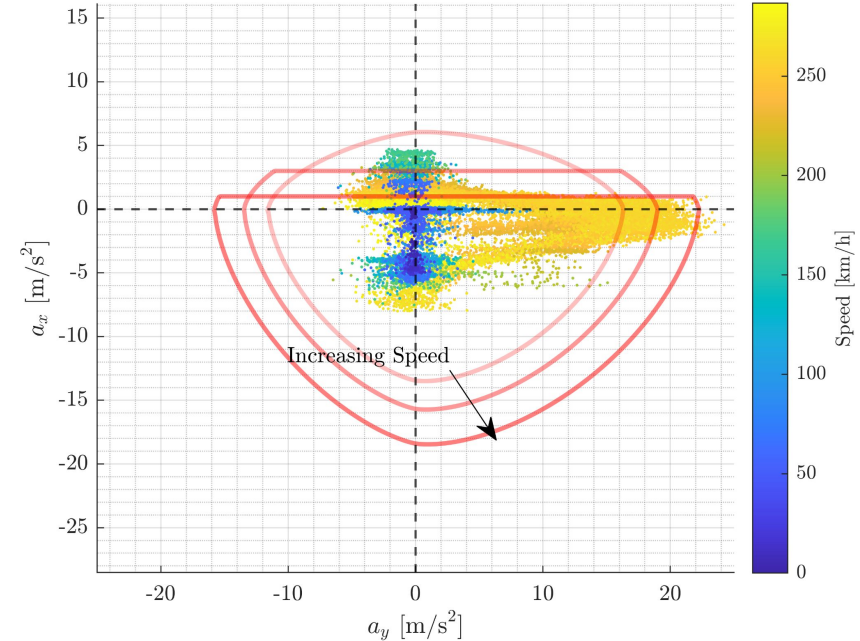
Global Planning - Lap-time Optimization

- Optimal Global Trajectory for Minimum Lap-Time: Optimization in Julia using a single/double track model
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Optimization-Based Trajectory Planning

- Model Predictive Control (MPC) framework.
- Planner serves as a Reference Governor for the motion controller.
- Constraints as a comprehensive approach to enforce safety and vehicle handling.



GG-V diagrams incorporating track banking (data from the IAC at IMS, 2024).

Multi-Planner Scheme for Overtakes

Drivable Tunnel Generator:

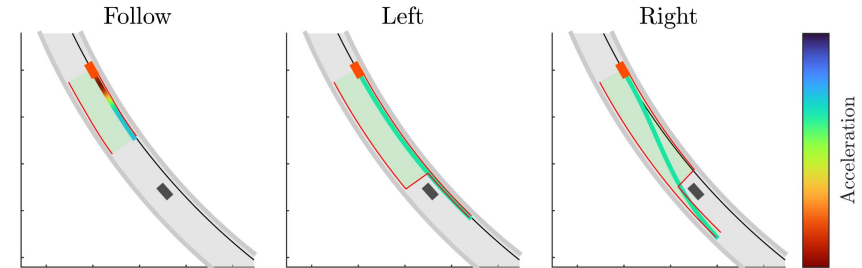
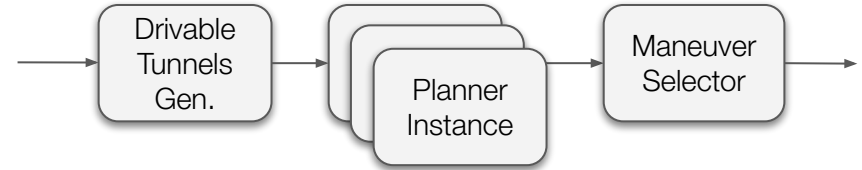
- Convex constraint subsets of the drivable space.
- Rule-based (e.g., right-of-way).

Multi-Planner Scheme:

- Multiple planner instances for different maneuvers (e.g., overtake left/right, follow).

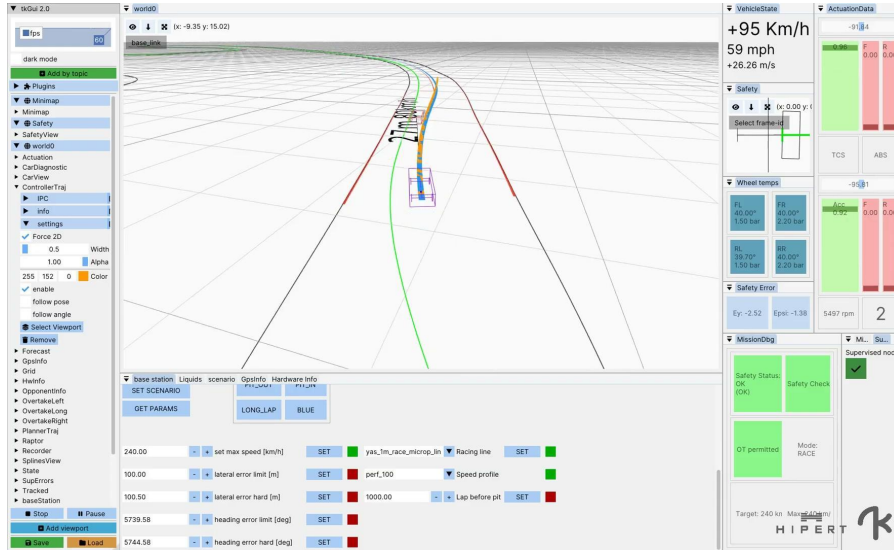
Decision Making:

- Selects the “best” maneuver among the available options.



Drivable tunnels and trajectory planning in three different maneuvers.

Local Planning - Overtakes



FMU-based Simulation at Yas Marina Circuit

- Path tracking and Planning with Model Predictive Control (MPC).
- Hierarchical approach
- Tube-based Overtake Planning
 - Comparing ego prediction with opponents forecastings
 - Decision-making with heuristics
 - Rule-compliant

Local Planning - Overtakes



Experimental testing at Las Vegas Motor Speedway

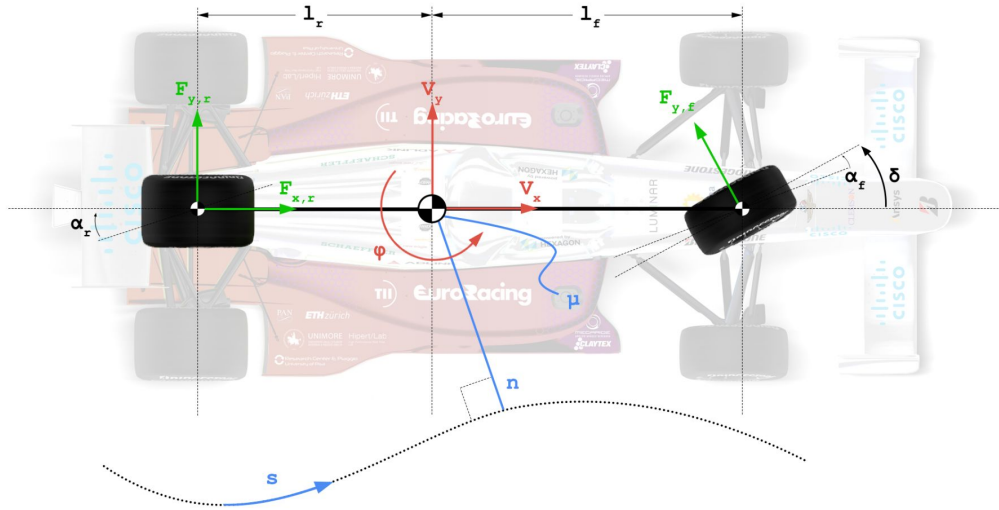
[submitted paper](#)

Motion Control

Control - Enhanced Bicycle Model

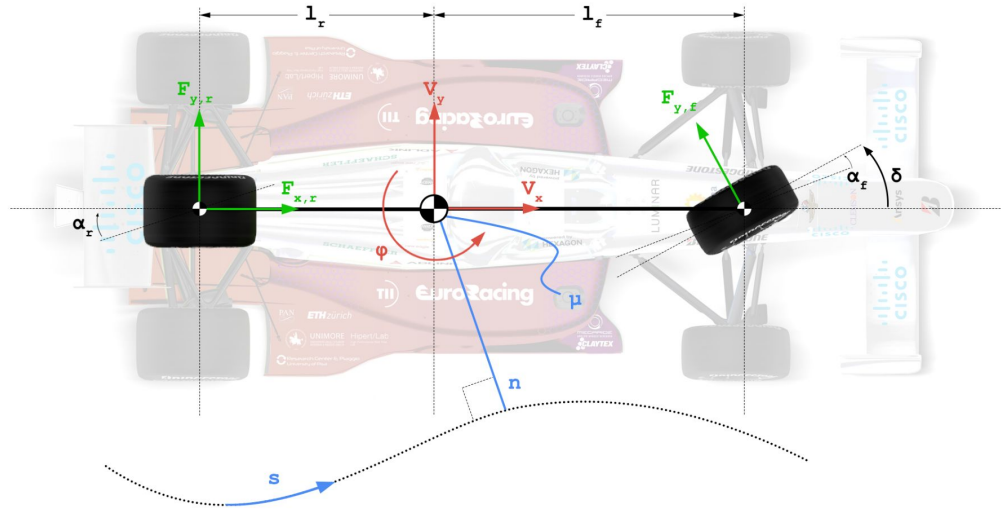
Formulation developed for considering:

- Banking
- Tires asymmetry
- Internal Combustion Engine
- Differential effects
- Combined Slip effects



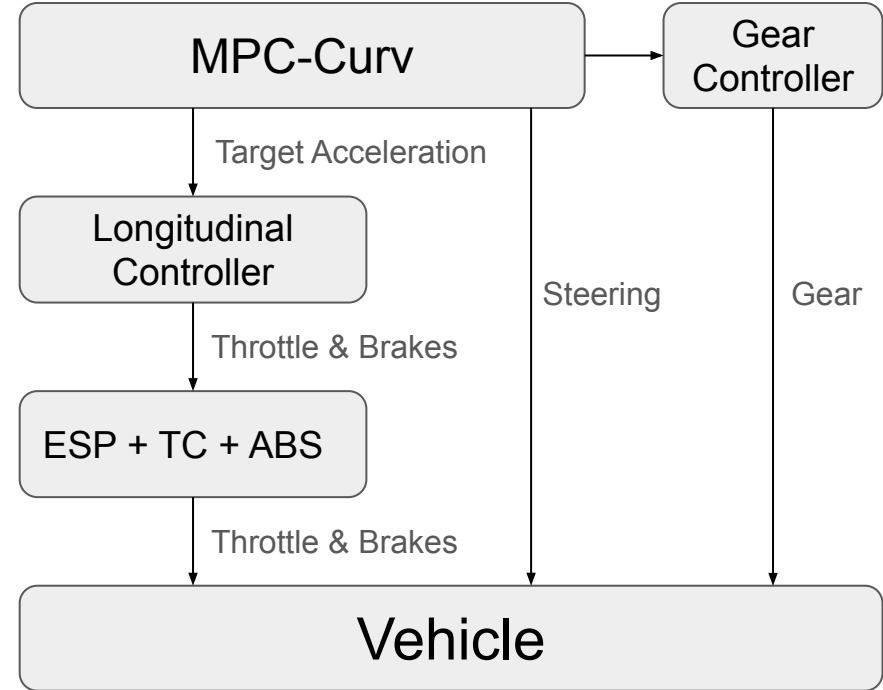
Control - Enhanced Bicycle Model

$$\begin{aligned}\dot{s} &= \frac{v_x \cos(\mu) - v_y \sin(\mu)}{1 - n \kappa(s)} \\ \dot{n} &= v_x \sin(\mu) + v_y \cos(\mu) \\ \dot{\mu} &= \dot{\phi} - \kappa(s) \dot{s} \\ \dot{v}_x &= \frac{1}{m} (F_{x_r} - F_d - F_{y_f} \sin(\delta) + F_{x_f} \cos(\delta) - F_{b_x} + m v_y \dot{\phi}) \\ \dot{v}_y &= \frac{1}{m} (F_{y_r} + F_{y_f} \cos(\delta) + F_{x_f} \sin(\delta) - F_{b_y} - m v_x \dot{\phi}) \\ \ddot{\phi} &= \frac{1}{I_z} (l_f (F_{y_f} \cos(\delta) + F_{x_f} \sin(\delta)) - l_r F_{y_r}) \\ \dot{\delta} &= \Delta \delta \\ \dot{T} &= \Delta T \\ \dot{B} &= \Delta B\end{aligned}$$

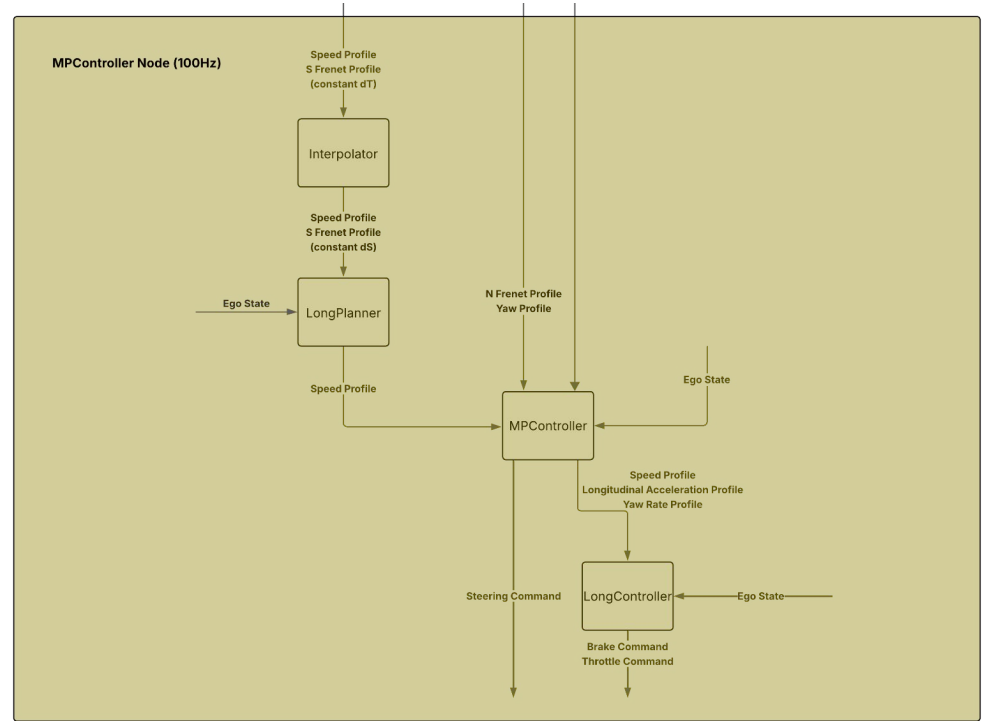
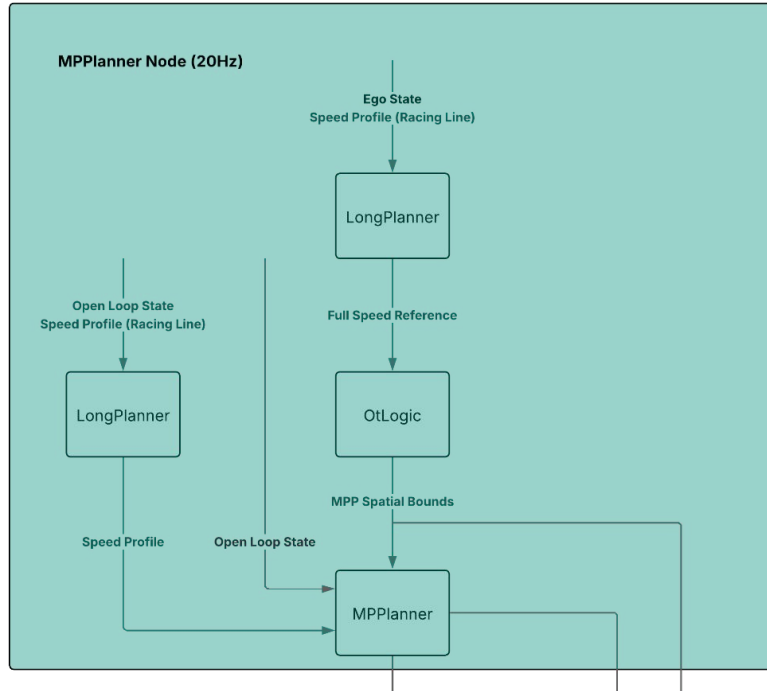


Motion and Vehicle Stability Control

- MPC-Curv
- Longitudinal Controller: Feedforward + PID
- Predictive Gear Controller
- Lower level controller
 - Traction Control
 - Active Braking System
 - Electronic Stability Control
- Emergency Controller:
 - Uses RADAR-based wall detection to pull over safely in case of localization failures.



Motion Planning and Control - full scheme



WHAT COULD POSSIBLY GO WRONG?

Many things can happen



- Edge cases not properly considered
- Bugs in the code
- Sensors faults and bugs
- Car's electronics and/or mechanics failures
- ...



Contacts and Events



Our Last Event

- Modena 09/06/25, Motor valley Fest
- Marzaglia track record (see onboard beside)

Next Event

- MIMO Monza 27-29/06/25

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Unimore Racing



unimoreracing

Research - References

- A. Raji et al., "[Motion Planning and Control for Multi Vehicle Autonomous Racing at High Speeds.](#)" 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)
- A. Raji et al., "[A Tricycle Model to Accurately Control an Autonomous Racecar with Locked Differential.](#)" 2023 IEEE 11th International Conference on Systems and Control (ICSC)
- A. Raji et al. "[er.autopilot 1.1: A Software Stack for Autonomous Racing on Oval and Road Course Tracks](#)", 2024, Field Robotics, 4, 99–137.
- A. Toschi et al., "[Guess the Drift with LOP-UKF: LiDAR Odometry and Pacejka Model for Real-Time Racecar Sideslip Estimation.](#)" 2024 IEEE Intelligent Vehicles Symposium (IV)
- F. Prignoli et al., "[RADAR-Based Safe Pull-Over of Autonomous Racing Cars in Localization Failure Scenarios](#)" 2024 European Control Conference (ECC)
- A. Raji, "[Model Predictive Planning and Control for Autonomous Racing, from HPC to Embedded Platforms](#)", PhD Thesis, Unipr, 2024
- A. Remonda et al. "[A Simulation Benchmark for Autonomous Racing with Large-Scale Human Data](#)", NeurIPS, 2024